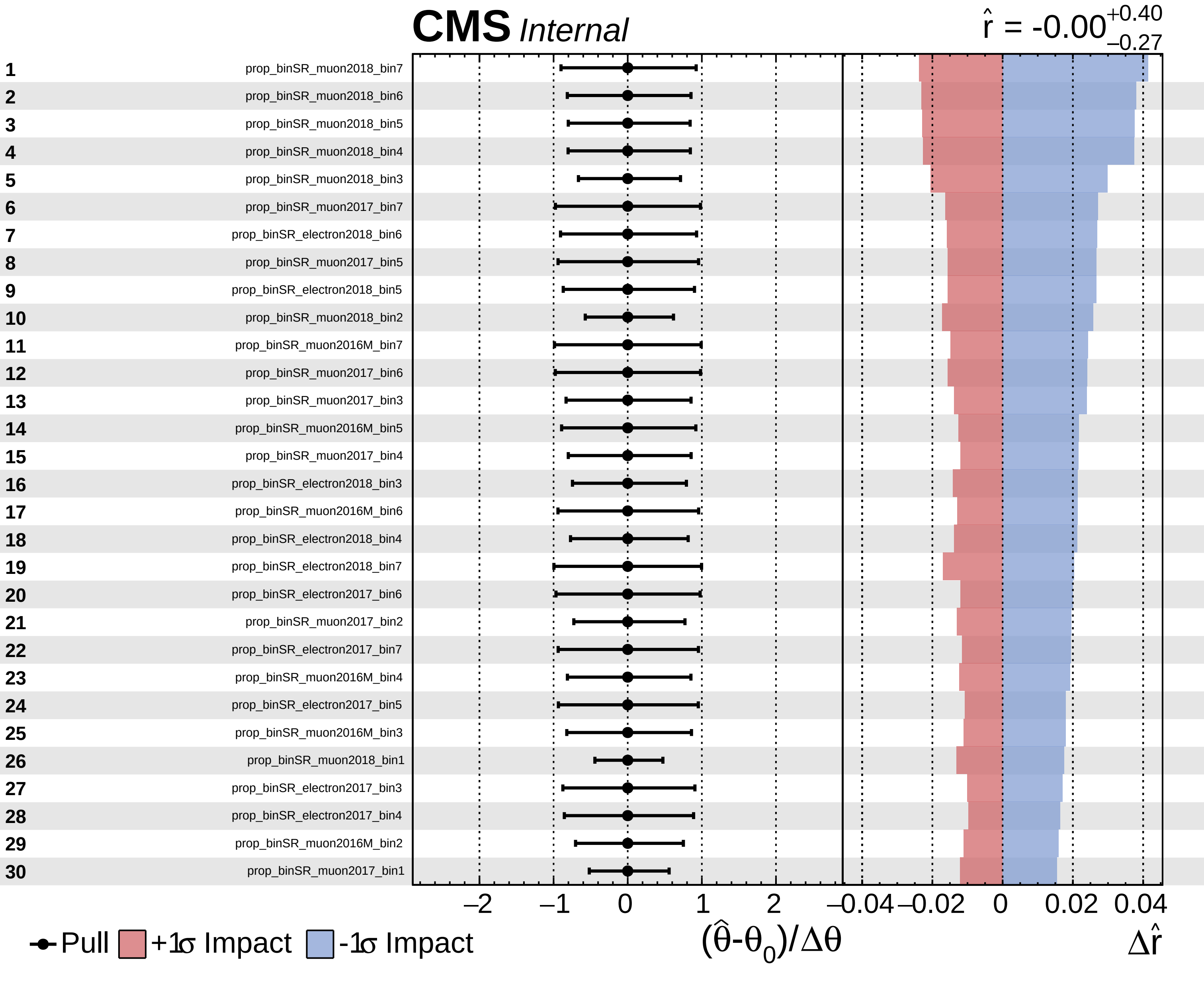
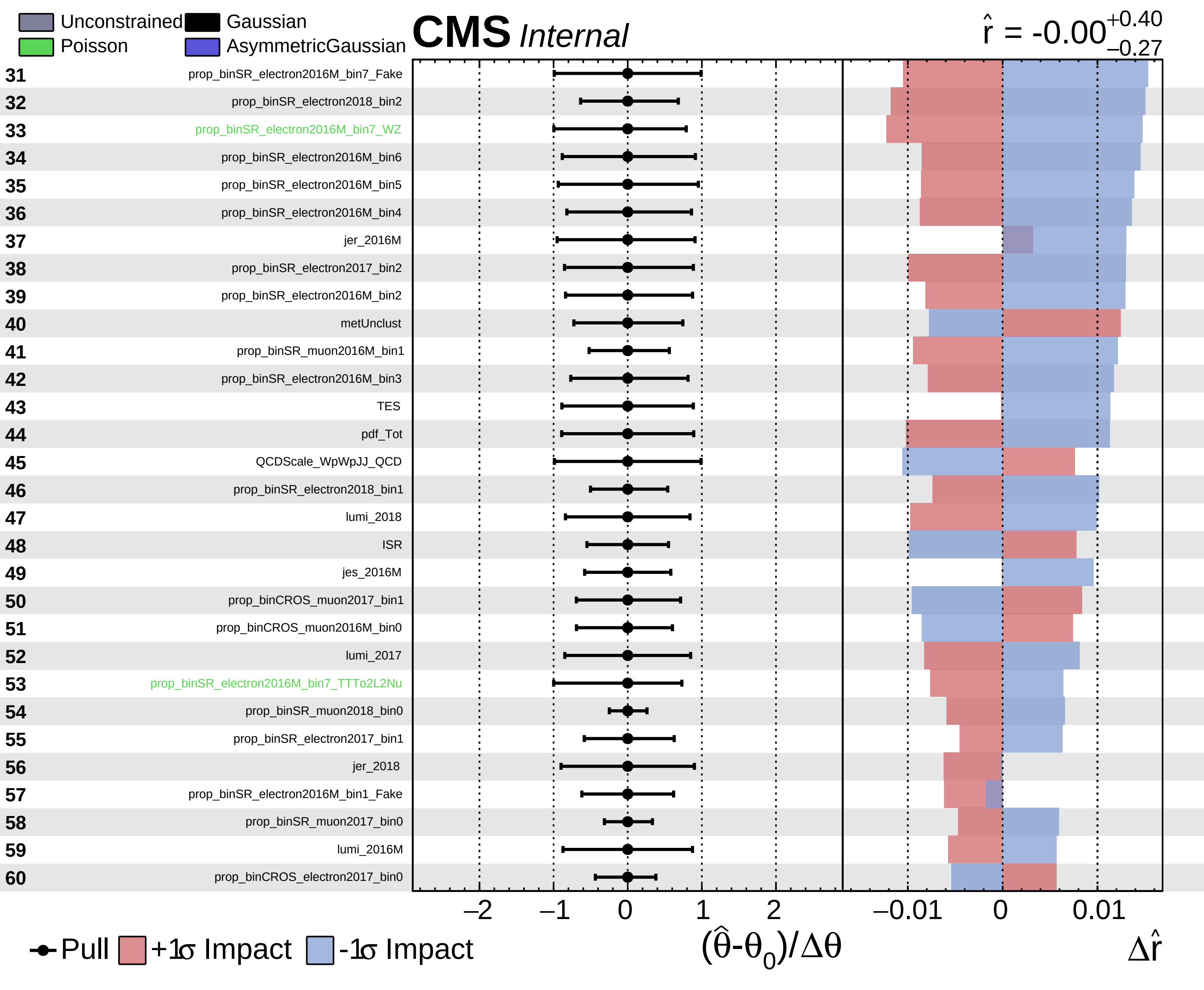
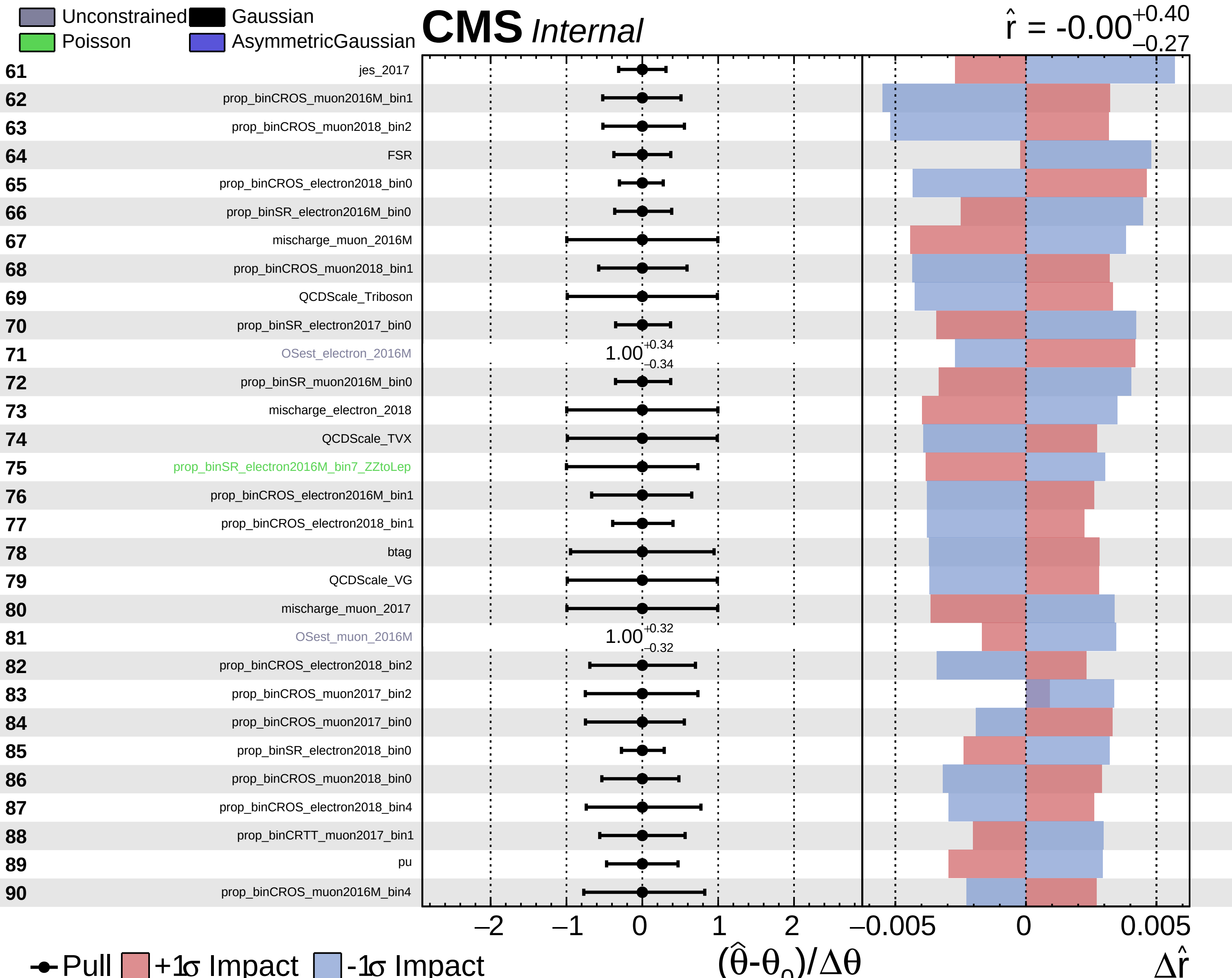


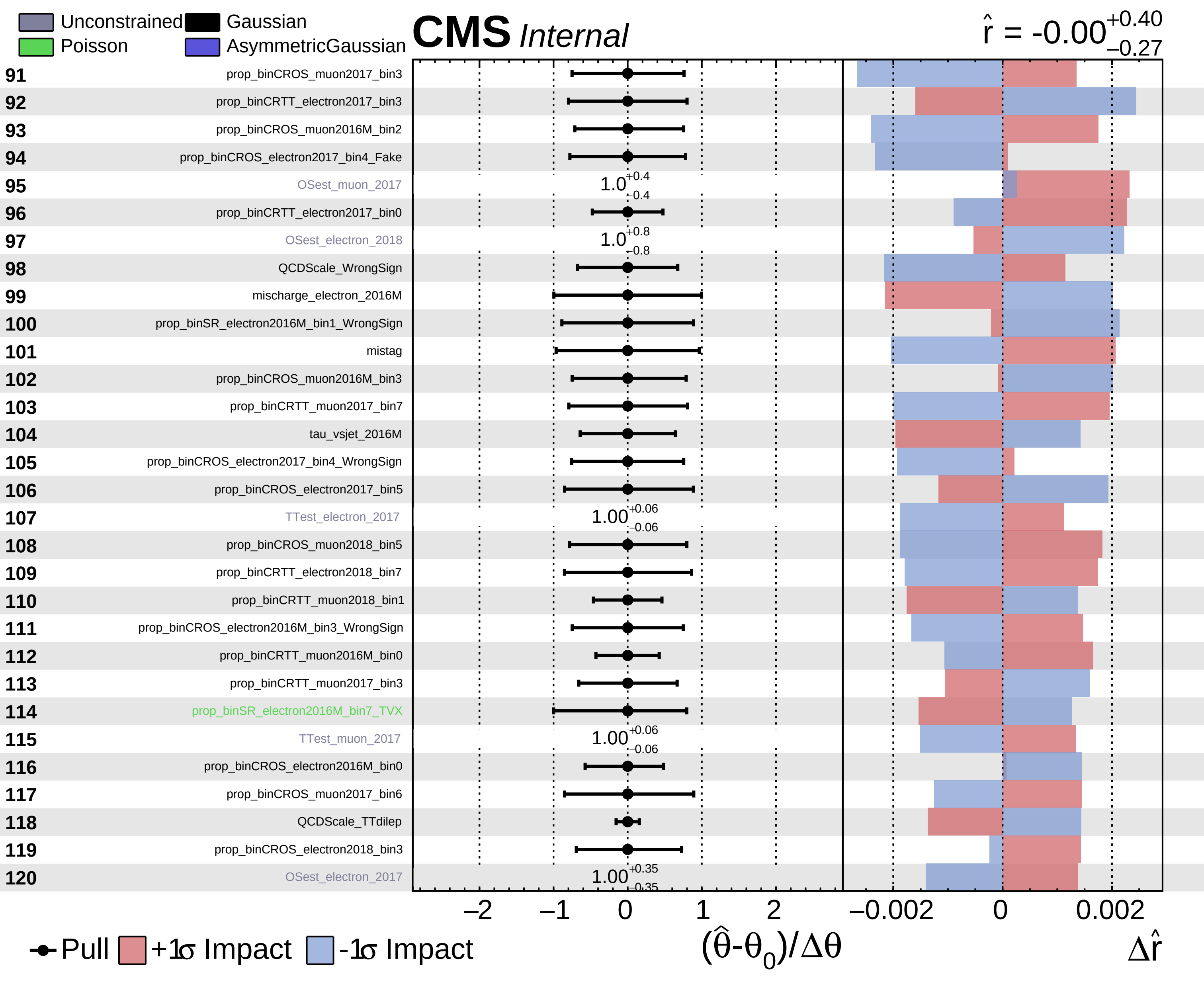
CMS Internal
$\hat{r} = -0.00^{+0.40}_{-0.27}$
1 prop_binSR_muon2018_bin7
2 prop_binSR_muon2018_bin6
3 prop_binSR_muon2018_bin5
4 prop_binSR_muon2018_bin4
5 prop_binSR_muon2018_bin3
6 prop_binSR_muon2017_bin7
7 prop_binSR_electron2018_bin6
8 prop_binSR_muon2017_bin5
9 prop_binSR_electron2018_bin5
10 prop_binSR_muon2018_bin2
11 prop_binSR_muon2016M_bin7
12 prop_binSR_muon2017_bin6
13 prop_binSR_muon2017_bin3
14 prop_binSR_muon2016M_bin5
15 prop_binSR_muon2017_bin4
16 prop_binSR_electron2018_bin3
17 prop_binSR_muon2016M_bin6
18 prop_binSR_electron2018_bin4
19 prop_binSR_electron2018_bin7
20 prop_binSR_electron2017_bin6
21 prop_binSR_muon2017_bin2
22 prop_binSR_electron2017_bin7
23 prop_binSR_muon2016M_bin4
24 prop_binSR_electron2017_bin5
25 prop_binSR_muon2016M_bin3
26 prop_binSR_muon2018_bin1
27 prop_binSR_electron2017_bin3
28 prop_binSR_electron2017_bin4
29 prop_binSR_muon2016M_bin2
30 prop_binSR_muon2017_bin1
−2 −1 0 1 2
−0.04 −0.02 0 0.02 0.04
Pull
+1σ Impact
-1σ Impact
$(\hat{\theta}-\theta_0)/\Delta\theta$
$\Delta\hat{r}$



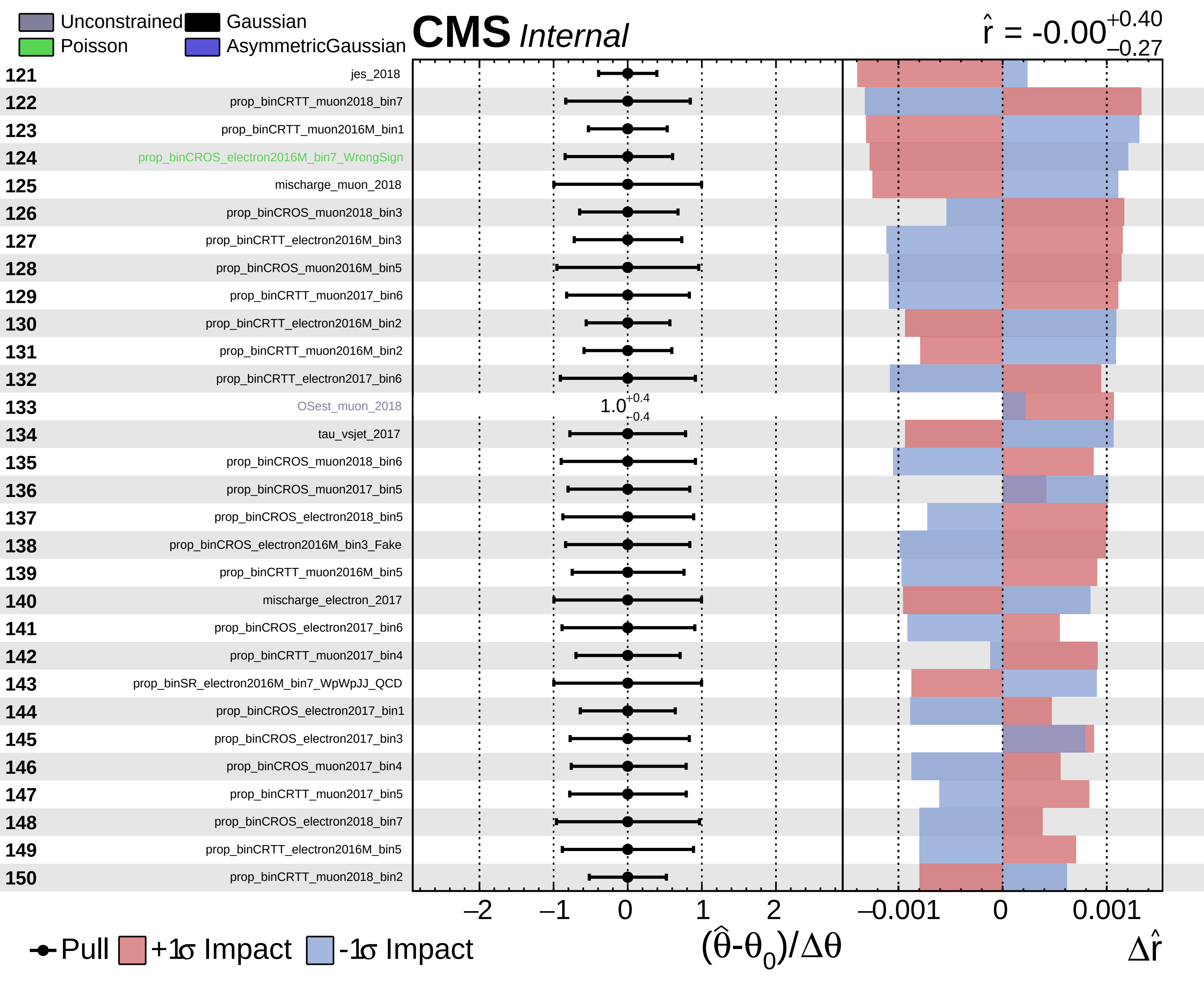
Unconstrained
Gaussian
Poisson
AsymmetricGaussian
CMS Internal
$\hat{r} = -0.00^{+0.40}_{-0.27}$
31 prop_binSR_electron2016M_bin7_Fake
32 prop_binSR_electron2018_bin2
33 prop_binSR_electron2016M_bin7_WZ
34 prop_binSR_electron2016M_bin6
35 prop_binSR_electron2016M_bin5
36 prop_binSR_electron2016M_bin4
37 jer_2016M
38 prop_binSR_electron2017_bin2
39 prop_binSR_electron2016M_bin2
40 metUnclust
41 prop_binSR_muon2016M_bin1
42 prop_binSR_electron2016M_bin3
43 TES
44 pdf_Tot
45 QCDScale_WpWpJJ_QCD
46 prop_binSR_electron2018_bin1
47 lumi_2018
48 ISR
49 jes_2016M
50 prop_binCROS_muon2017_bin1
51 prop_binCROS_muon2016M_bin0
52 lumi_2017
53 prop_binSR_electron2016M_bin7_TTTo2L2Nu
54 prop_binSR_muon2018_bin0
55 prop_binSR_electron2017_bin1
56 jer_2018
57 prop_binSR_electron2016M_bin1_Fake
58 prop_binSR_muon2017_bin0
59 lumi_2016M
60 prop_binCROS_electron2017_bin0
−2
−1
0
1
2
−0.01
0
0.01
Pull
+1σ Impact
-1σ Impact
$(\hat{\theta}-\theta_0)/\Delta\theta$
$\Delta\hat{r}$



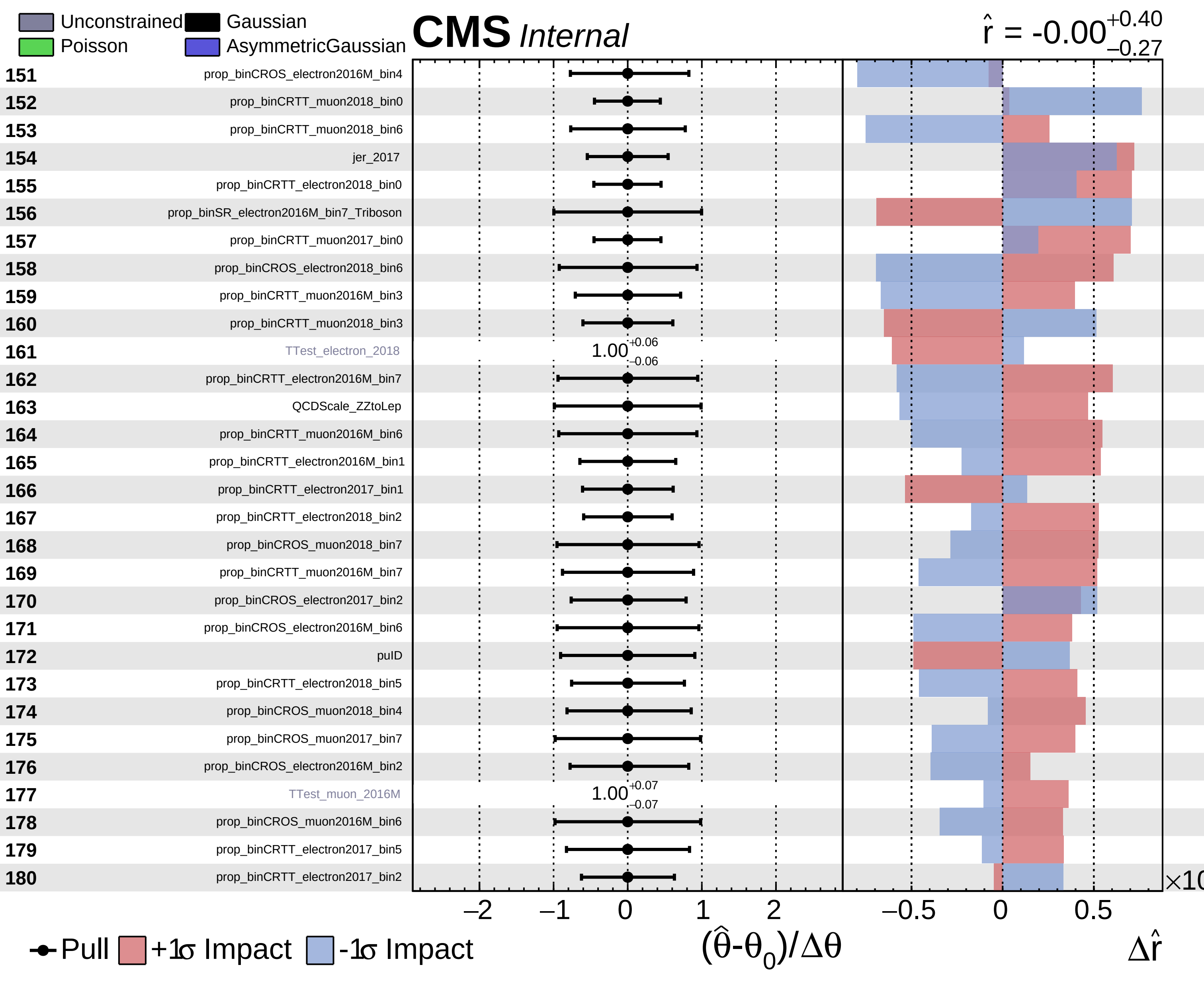
Unconstrained
Gaussian
Poisson
AsymmetricGaussian
CMS Internal
$\hat{r} = -0.00^{+0.40}_{-0.27}$
61 jes_2017
62 prop_binCROS_muon2016M_bin1
63 prop_binCROS_muon2018_bin2
64 FSR
65 prop_binCROS_electron2018_bin0
66 prop_binSR_electron2016M_bin0
67 mischarge_muon_2016M
68 prop_binCROS_muon2018_bin1
69 QCDScale_Triboson
70 prop_binSR_electron2017_bin0
71 OSest_electron_2016M
$1.00^{+0.34}_{-0.34}$
72 prop_binSR_muon2016M_bin0
73 mischarge_electron_2018
74 QCDScale_TVX
75 prop_binSR_electron2016M_bin7_ZZtoLep
76 prop_binCROS_electron2016M_bin1
77 prop_binCROS_electron2018_bin1
78 btag
79 QCDScale_VG
80 mischarge_muon_2017
81 OSest_muon_2016M
$1.00^{+0.32}_{-0.32}$
82 prop_binCROS_electron2018_bin2
83 prop_binCROS_muon2017_bin2
84 prop_binCROS_muon2017_bin0
85 prop_binSR_electron2018_bin0
86 prop_binCROS_muon2018_bin0
87 prop_binCROS_electron2018_bin4
88 prop_binCRTT_muon2017_bin1
89 pu
90 prop_binCROS_muon2016M_bin4
−2
−1
0
1
2
−0.005
0
0.005
Pull
+1σ Impact
-1σ Impact
$(\hat{\theta}-\theta_0)/\Delta\theta$
$\Delta\hat{r}$



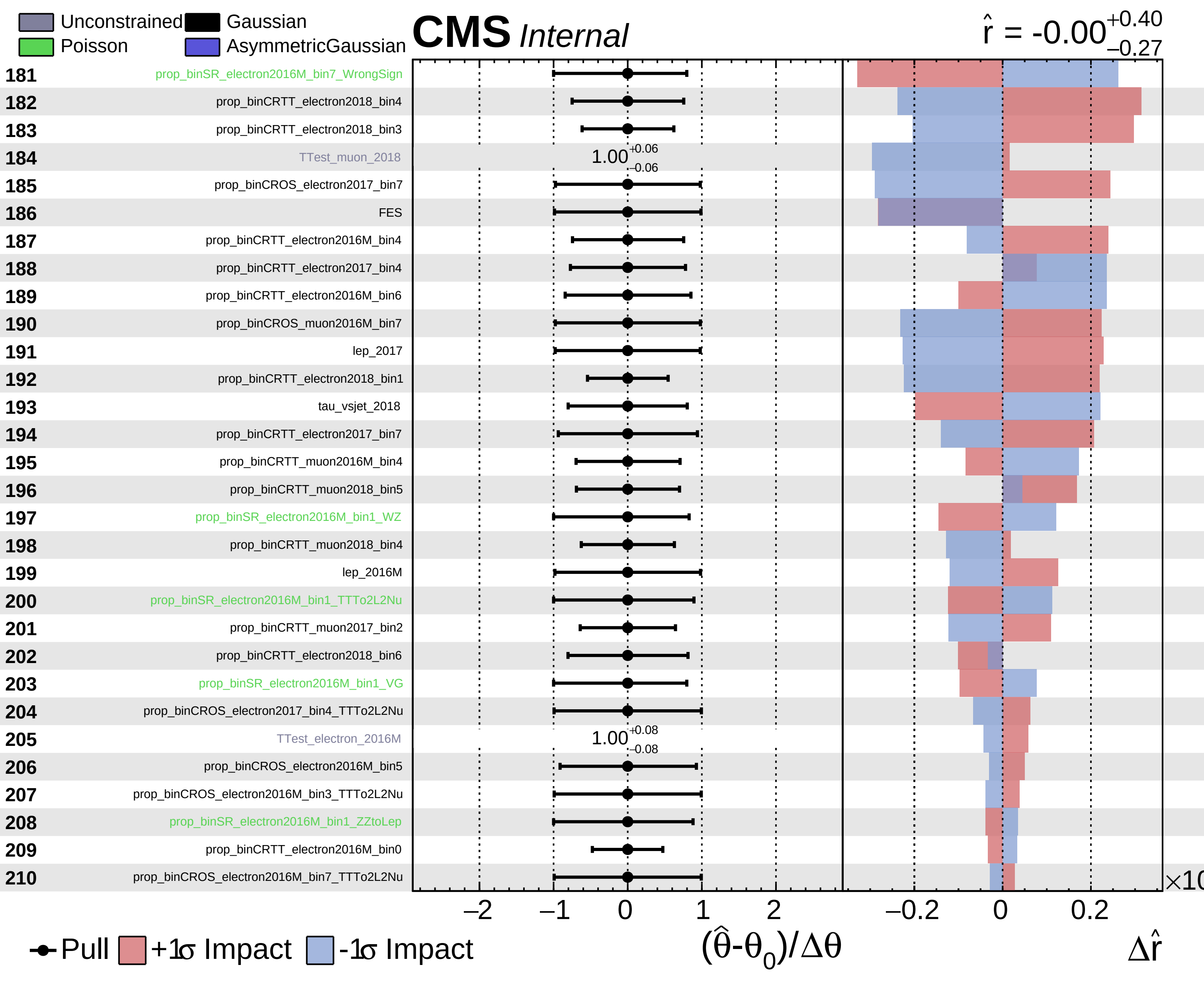
Unconstrained
Gaussian
Poisson
AsymmetricGaussian
CMS Internal
$\hat{r} = -0.00^{+0.40}_{-0.27}$
91 prop_binCROS_muon2017_bin3
92 prop_binCRTT_electron2017_bin3
93 prop_binCROS_muon2016M_bin2
94 prop_binCROS_electron2017_bin4_Fake
95 OSest_muon_2017
$1.0^{+0.4}_{-0.4}$
96 prop_binCRTT_electron2017_bin0
97 OSest_electron_2018
$1.0^{+0.8}_{-0.8}$
98 QCDScale_WrongSign
99 mischarge_electron_2016M
100 prop_binSR_electron2016M_bin1_WrongSign
101 mistag
102 prop_binCROS_muon2016M_bin3
103 prop_binCRTT_muon2017_bin7
104 tau_vsjet_2016M
105 prop_binCROS_electron2017_bin4_WrongSign
106 prop_binCROS_electron2017_bin5
107 TTest_electron_2017
$1.00^{+0.06}_{-0.06}$
108 prop_binCROS_muon2018_bin5
109 prop_binCRTT_electron2018_bin7
110 prop_binCRTT_muon2018_bin1
111 prop_binCROS_electron2016M_bin3_WrongSign
112 prop_binCRTT_muon2016M_bin0
113 prop_binCRTT_muon2017_bin3
114 prop_binSR_electron2016M_bin7_TVX
115 TTest_muon_2017
$1.00^{+0.06}_{-0.06}$
116 prop_binCROS_electron2016M_bin0
117 prop_binCROS_muon2017_bin6
118 QCDScale_TTdilep
119 prop_binCROS_electron2018_bin3
120 OSest_electron_2017
$1.00^{+0.35}_{-0.35}$
−2
−1
0
1
2
−0.002
0
0.002
Pull
+1σ Impact
-1σ Impact
$(\hat{\theta}-\theta_0)/\Delta\theta$
$\Delta\hat{r}$

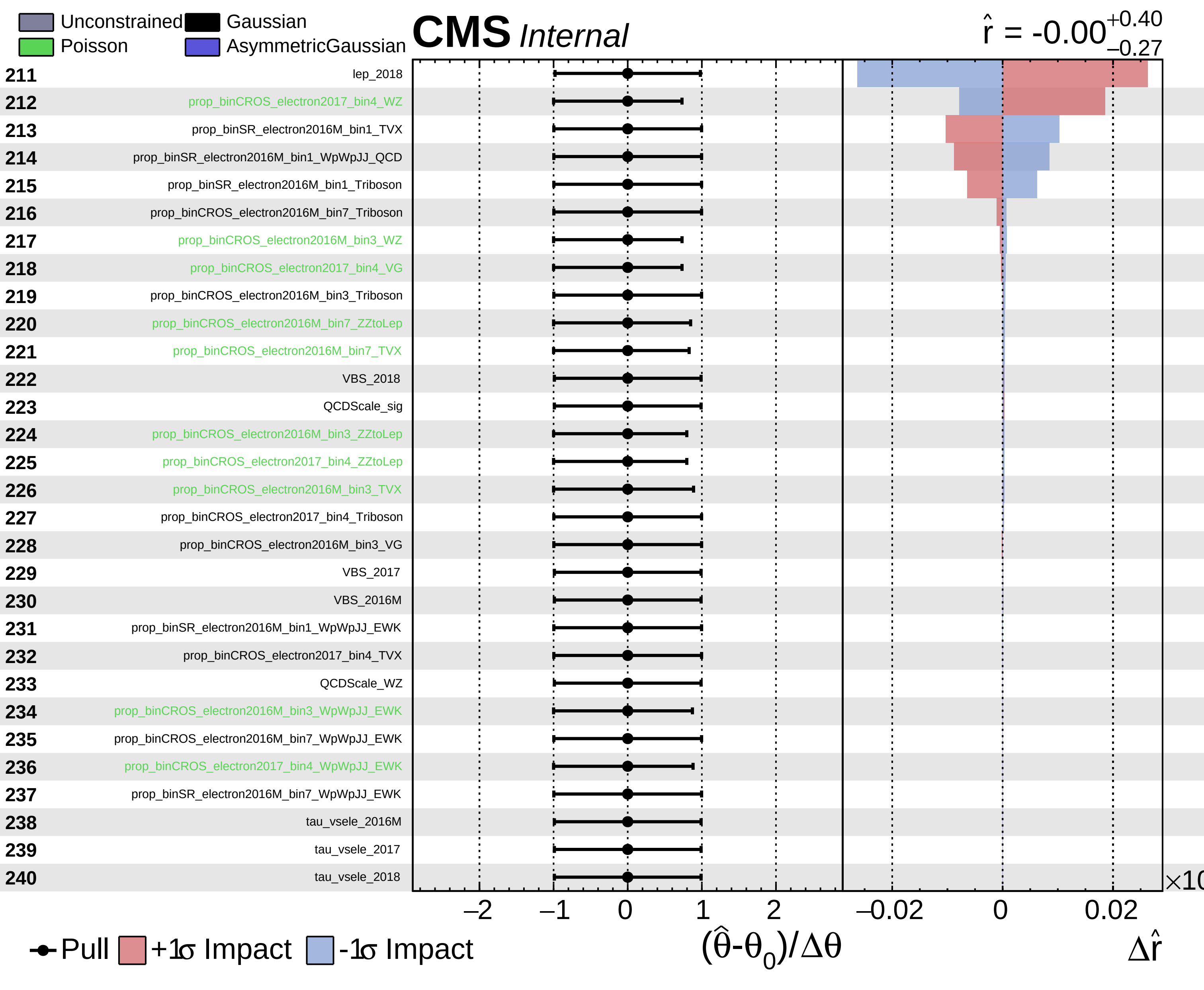


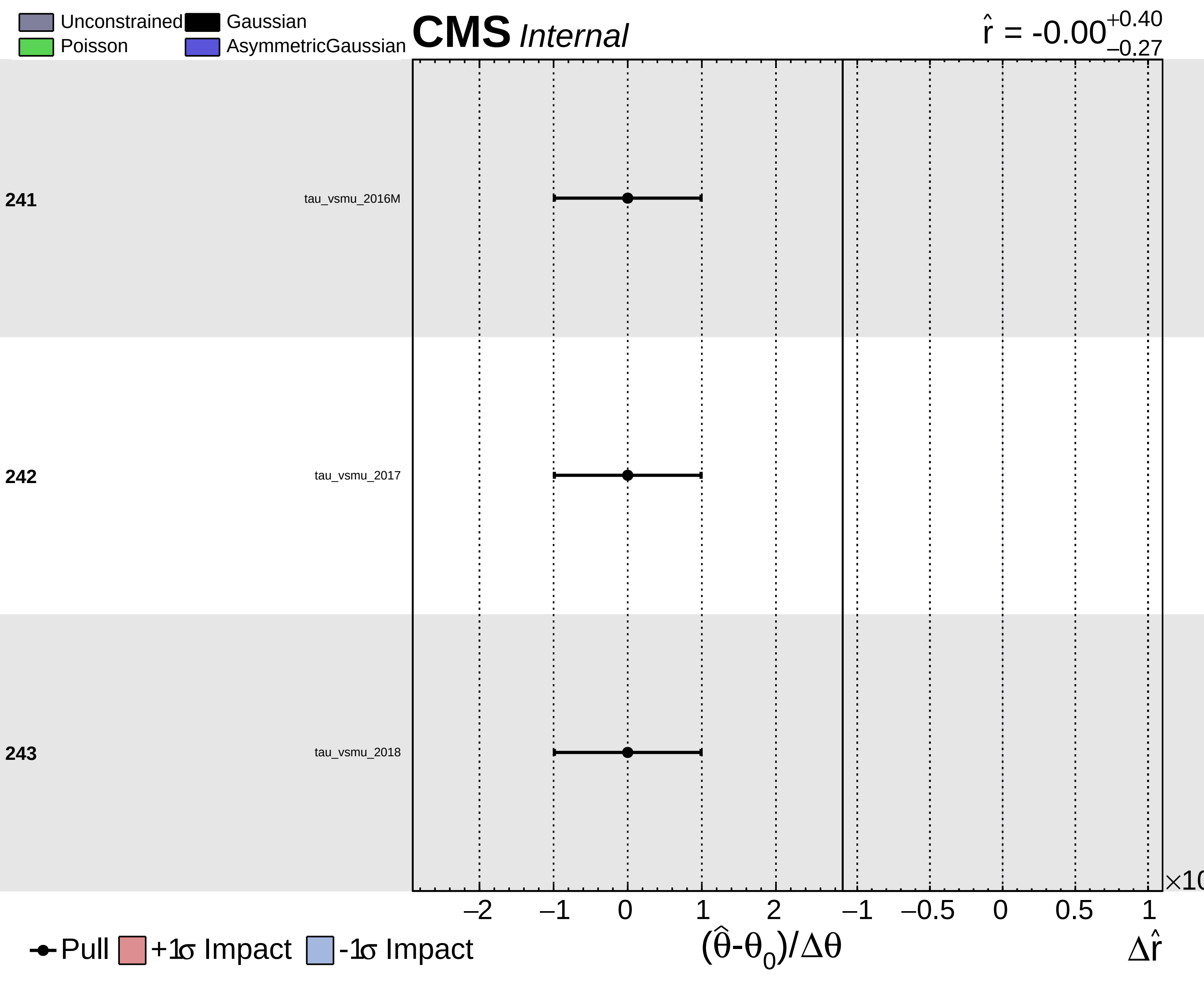
Unconstrained
Gaussian
Poisson
AsymmetricGaussian
CMS Internal
$\hat{r} = -0.00^{+0.40}_{-0.27}$
121 jes_2018
122 prop_binCRTT_muon2018_bin7
123 prop_binCRTT_muon2016M_bin1
124 prop_binCROS_electron2016M_bin7_WrongSign
125 mischarge_muon_2018
126 prop_binCROS_muon2018_bin3
127 prop_binCRTT_electron2016M_bin3
128 prop_binCROS_muon2016M_bin5
129 prop_binCRTT_muon2017_bin6
130 prop_binCRTT_electron2016M_bin2
131 prop_binCRTT_muon2016M_bin2
132 prop_binCRTT_electron2017_bin6
133 OSest_muon_2018
$1.0^{+0.4}_{-0.4}$
134 tau_vsjet_2017
135 prop_binCROS_muon2018_bin6
136 prop_binCROS_muon2017_bin5
137 prop_binCROS_electron2018_bin5
138 prop_binCROS_electron2016M_bin3_Fake
139 prop_binCRTT_muon2016M_bin5
140 mischarge_electron_2017
141 prop_binCROS_electron2017_bin6
142 prop_binCRTT_muon2017_bin4
143 prop_binSR_electron2016M_bin7_WpWpJJ_QCD
144 prop_binCROS_electron2017_bin1
145 prop_binCROS_electron2017_bin3
146 prop_binCROS_muon2017_bin4
147 prop_binCRTT_muon2017_bin5
148 prop_binCROS_electron2018_bin7
149 prop_binCRTT_electron2016M_bin5
150 prop_binCRTT_muon2018_bin2
−2
−1
0
1
2
−0.001
0
0.001
Pull
+1σ Impact
-1σ Impact
$(\hat{\theta}-\theta_0)/\Delta\theta$
$\Delta\hat{r}$



Unconstrained
Gaussian
Poisson
AsymmetricGaussian
CMS Internal
$\hat{r} = -0.00^{+0.40}_{-0.27}$
151 prop_binCROS_electron2016M_bin4
152 prop_binCRTT_muon2018_bin0
153 prop_binCRTT_muon2018_bin6
154 jer_2017
155 prop_binCRTT_electron2018_bin0
156 prop_binSR_electron2016M_bin7_Triboson
157 prop_binCRTT_muon2017_bin0
158 prop_binCROS_electron2018_bin6
159 prop_binCRTT_muon2016M_bin3
160 prop_binCRTT_muon2018_bin3
161 TTest_electron_2018
$1.00^{+0.06}_{-0.06}$
162 prop_binCRTT_electron2016M_bin7
163 QCDScale_ZZtoLep
164 prop_binCRTT_muon2016M_bin6
165 prop_binCRTT_electron2016M_bin1
166 prop_binCRTT_electron2017_bin1
167 prop_binCRTT_electron2018_bin2
168 prop_binCROS_muon2018_bin7
169 prop_binCRTT_muon2016M_bin7
170 prop_binCROS_electron2017_bin2
171 prop_binCROS_electron2016M_bin6
172 puID
173 prop_binCRTT_electron2018_bin5
174 prop_binCROS_muon2018_bin4
175 prop_binCROS_muon2017_bin7
176 prop_binCROS_electron2016M_bin2
177 TTest_muon_2016M
$1.00^{+0.07}_{-0.07}$
178 prop_binCROS_muon2016M_bin6
179 prop_binCRTT_electron2017_bin5
180 prop_binCRTT_electron2017_bin2
−2 −1 0 1 2
−0.5 0 0.5
×10
Pull
+1σ Impact
-1σ Impact
$(\hat{\theta}-\theta_0)/\Delta\theta$
$\Delta\hat{r}$


Unconstrained
Gaussian
Poisson
AsymmetricGaussian
CMS Internal
$\hat{r} = -0.00^{+0.40}_{-0.27}$
181 prop_binSR_electron2016M_bin7_WrongSign
182 prop_binCRTT_electron2018_bin4
183 prop_binCRTT_electron2018_bin3
184 TTest_muon_2018
$1.00^{+0.06}_{-0.06}$
185 prop_binCROS_electron2017_bin7
186 FES
187 prop_binCRTT_electron2016M_bin4
188 prop_binCRTT_electron2017_bin4
189 prop_binCRTT_electron2016M_bin6
190 prop_binCROS_muon2016M_bin7
191 lep_2017
192 prop_binCRTT_electron2018_bin1
193 tau_vsjet_2018
194 prop_binCRTT_electron2017_bin7
195 prop_binCRTT_muon2016M_bin4
196 prop_binCRTT_muon2018_bin5
197 prop_binSR_electron2016M_bin1_WZ
198 prop_binCRTT_muon2018_bin4
199 lep_2016M
200 prop_binSR_electron2016M_bin1_TTTo2L2Nu
201 prop_binCRTT_muon2017_bin2
202 prop_binCRTT_electron2018_bin6
203 prop_binSR_electron2016M_bin1_VG
204 prop_binCROS_electron2017_bin4_TTTo2L2Nu
205 TTest_electron_2016M
$1.00^{+0.08}_{-0.08}$
206 prop_binCROS_electron2016M_bin5
207 prop_binCROS_electron2016M_bin3_TTTo2L2Nu
208 prop_binSR_electron2016M_bin1_ZZtoLep
209 prop_binCRTT_electron2016M_bin0
210 prop_binCROS_electron2016M_bin7_TTTo2L2Nu
−2
−1
0
1
2
−0.2
0
0.2
×10
Pull
+1σ Impact
-1σ Impact
$(\hat{\theta}-\theta_0)/\Delta\theta$
$\Delta\hat{r}$


Unconstrained
Gaussian
Poisson
AsymmetricGaussian
CMS Internal
$\hat{r} = -0.00^{+0.40}_{-0.27}$
211 lep_2018
212 prop_binCROS_electron2017_bin4_WZ
213 prop_binSR_electron2016M_bin1_TVX
214 prop_binSR_electron2016M_bin1_WpWpJJ_QCD
215 prop_binSR_electron2016M_bin1_Triboson
216 prop_binCROS_electron2016M_bin7_Triboson
217 prop_binCROS_electron2016M_bin3_WZ
218 prop_binCROS_electron2017_bin4_VG
219 prop_binCROS_electron2016M_bin3_Triboson
220 prop_binCROS_electron2016M_bin7_ZZtoLep
221 prop_binCROS_electron2016M_bin7_TVX
222 VBS_2018
223 QCDScale_sig
224 prop_binCROS_electron2016M_bin3_ZZtoLep
225 prop_binCROS_electron2017_bin4_ZZtoLep
226 prop_binCROS_electron2016M_bin3_TVX
227 prop_binCROS_electron2017_bin4_Triboson
228 prop_binCROS_electron2016M_bin3_VG
229 VBS_2017
230 VBS_2016M
231 prop_binSR_electron2016M_bin1_WpWpJJ_EWK
232 prop_binCROS_electron2017_bin4_TVX
233 QCDScale_WZ
234 prop_binCROS_electron2016M_bin3_WpWpJJ_EWK
235 prop_binCROS_electron2016M_bin7_WpWpJJ_EWK
236 prop_binCROS_electron2017_bin4_WpWpJJ_EWK
237 prop_binSR_electron2016M_bin7_WpWpJJ_EWK
238 tau_vsele_2016M
239 tau_vsele_2017
240 tau_vsele_2018
−2
−1
0
1
2
−0.02
0
0.02
×10
Pull
+1σ Impact
-1σ Impact
$(\hat{\theta}-\theta_0)/\Delta\theta$
$\Delta\hat{r}$



Unconstrained
Gaussian
Poisson
AsymmetricGaussian
CMS Internal
$\hat{r} = -0.00^{+0.40}_{-0.27}$
241
tau_vsmu_2016M
242
tau_vsmu_2017
243
tau_vsmu_2018
−2
−1
0
1
2
−1
−0.5
0
0.5
1
×10
Pull
+1σ Impact
-1σ Impact
$(\hat{\theta}-\theta_0)/\Delta\theta$
$\Delta\hat{r}$